A HALF-DAY SHORT COURSE

Time Series Forecasting

From VAR to Transformers and Diffusion

Attention · Patching · Foundation Models · Denoising Diffusion

Course Roadmap (about 4.5 hours)

0:00 – 0:50	**Module 1 — Classical Multivariate Models**	VAR/VARMA, impulse response, Granger causality, cointegration & ECM
0:50 – 1:50	**Module 2 — The Transformer Era**	From VAR to attention, attention math, Informer to PatchTST
1:50 – 2:45	**Module 3 — Foundation Models**	Zero-shot forecasting, tokenizing time, the 2026 model zoo, live demo
2:45 – 2:55	**Break**	Coffee
2:55 – 3:55	**Module 4 — Diffusion Models**	Gaussian densities to scenarios, TimeGrad, CSDI/SSSD, the frontier
3:55 – 4:45	**Module 5 — Hands-On & Practice**	Six coding exercises, evaluation, model selection, open problems

What You Will Be Able to Do

1. **Specify and estimate** classical multivariate models — VAR/VARMA — and read impulse-response functions, Granger-causality tests, and cointegration/ECM.
2. **Formulate** point and probabilistic forecasting problems and pick the right metrics (MASE, CRPS, quantile loss).
3. **Explain** self-attention and how transformers were adapted to time series — and why a linear model briefly beat them all.
4. **Use** zero-shot foundation models (Chronos-2, TimesFM 2.5, Moirai-2) and know when they are the right tool.
5. **Derive** the intuition behind denoising diffusion and map it to forecasters such as TimeGrad, CSDI, and TSDiff.
6. **Decide** which model class fits a given scenario, and backtest it without fooling yourself.

MODULE 1 **0:00 – 0:50**

Classical Multivariate Models

VAR & VARMA · impulse response · Granger causality · cointegration & ECM

From Univariate AR to VAR

A univariate AR(p) predicts a series from its own past:

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t.$$

But series rarely move alone — interest rates, demand, and prices co-evolve.

A **vector autoregression** lets every variable depend on the recent past of *all* variables, capturing feedback and co-movement.

Why multivariate?

- cross-series predictive power (does x help forecast y?),
- shock propagation through a system,
- long-run equilibria between series.

These three questions map onto the three tools in this module: Granger causality, impulse response, and cointegration.

VAR(p): Definition and Estimation

For an $n \times 1$ vector y_t (zero mean):

$$y_t = A_1 y_{t-1} + \cdots + A_p y_{t-p} + \varepsilon_t, \qquad \varepsilon_t \sim \text{iid}\, \mathcal{N}(0, W).$$

Estimation

- Each equation has the *same* regressors (the stacked lags) $\Rightarrow$ estimate **equation-by-equation OLS**.
- W may be non-diagonal: shocks are contemporaneously correlated.

Stability / stationarity

- Stable if all roots of $\det(I - A_1 z - \cdots - A_p z^p) = 0$ lie *outside* the unit circle.
- Then a vector MA(∞) representation exists — the basis for impulse response.

Lab preview: in Exercise 2 you simulate a VAR(1), recover A by OLS ($\hat{A} \approx A$), and read off its impulse responses.

VARMA Models

Adding moving-average terms gives the general **VARMA(p, q)**:

$$y_t = \underbrace{A_1 y_{t-1} + \cdots + A_p y_{t-p}}_{\text{autoregressive}} + \varepsilon_t + \underbrace{B_1 \varepsilon_{t-1} + \cdots + B_q \varepsilon_{t-q}}_{\text{moving average}}.$$

- $\varepsilon_t \sim \text{iid}\, \mathcal{N}(0, W)$, with W possibly non-diagonal (contemporaneous correlation).
- Assumes *no* serial correlation in residuals and conditional homoskedasticity (no GARCH).
- More parsimonious than a long pure VAR, but estimation is harder (the MA part is nonlinear).

In practice a well-specified VAR(p) is often preferred for its simple OLS estimation; VARMA matters when the MA structure is strong.

Impulse-Response Analysis

Idea. How do shocks propagate? Invert a stable VAR into a vector MA(∞):

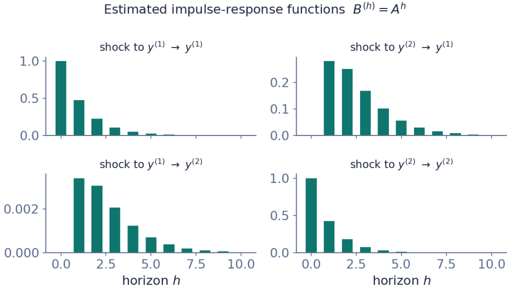
$$y_t = \varepsilon_t + B_1\varepsilon_{t-1} + B_2\varepsilon_{t-2} + \cdots$$

The matrices B_h are built recursively from the A_i.

The **impulse-response function** plots

$$B_{ij}^{(h)} = \text{effect on } y_{t+h}^{(i)} \text{ of a unit shock to } \varepsilon_t^{(j)},$$

tracing one variable's response to another's shock across the horizon h.



Estimated IRFs for a fitted VAR(1). Bottom-left is ≈ 0: a shock to $y^{(1)}$ barely moves $y^{(2)}$ — a preview of Granger non-causality.

Granger Causality

Question. Do past values of x_t help predict y_t, beyond y's own past?

$$y_t = a_1 y_{t-1} + \cdots + a_p y_{t-p} + b_1 x_{t-1} + \cdots + b_p x_{t-p} + u_t.$$

Null hypothesis (no causality $x \to y$):

$$H_0 : \; b_1 = b_2 = \cdots = b_p = 0.$$

Lab: on the simulated system you will find $F_{y_1 \to y_2} \approx 0$ (cannot reject H_0) but $F_{y_2 \to y_1} \gg 0$ — recovering the asymmetry built into A.

Tested with an F- (Wald) test on the joint restriction.

"Granger causality" is predictive, not structural — it says x helps forecast y, not that x *causes* y in a deeper sense.

Unit Roots, Cointegration, and the ECM

Unit roots. $y_t \sim I(1)$ if Δy_t is stationary. Regressing one $I(1)$ series on another risks *spurious regression*.

Cointegration. If $\mathbf{y}_t \sim I(1)$ but a linear combination $\boldsymbol{\beta}^\top \mathbf{y}_t$ is $I(0)$, the series share a long-run equilibrium; $\boldsymbol{\beta}$ is a cointegrating vector.

Error-correction model. A cointegrated system is a VAR in differences plus an equilibrium term:

$$\Delta \mathbf{y}_t = -\alpha \boldsymbol{\beta}^\top \mathbf{y}_{t-1} + \sum_j B_j \Delta \mathbf{y}_{t-j} + \boldsymbol{\varepsilon}_t.$$

- $\boldsymbol{\beta}$: cointegrating vectors (long run),
- α: speed of adjustment back to equilibrium.

Key implication: if $\mathbf{y}_t$ is cointegrated, a VAR in differences *alone* is mis-specified — you must keep the error-correction term. We revisit this when comparing to neural models, which do not enforce it.

Bridge: From VAR to Attention

Everything that follows generalizes the VAR you just saw.

Classical — VAR(p)

$$y_t = \sum_{i=1}^{p} A_i \, y_{t-i} + \varepsilon_t$$

- Coefficients A_i *fixed*, estimated by OLS.
- *Linear*, finite-lag combination of the past.

Modern — a transformer

$$\hat{y}_t = \sum_{i=1}^{L} \underbrace{a_i(y_{t-L:t-1})}_{\text{data-dependent}} \, v(y_{t-i})$$

- Weights a_i *recomputed for every window*.
- *Nonlinear* combination of learned values.

Attention weights play the role of the VAR coefficients A_i — but data-dependent instead of fixed. A causal mask keeps attention backward-looking, exactly like a VAR. **On to Module 2.**

MODULE 2 0:50 – 1:50

The Transformer Era

From VAR to attention · attention math · Informer to PatchTST

The Forecasting Problem

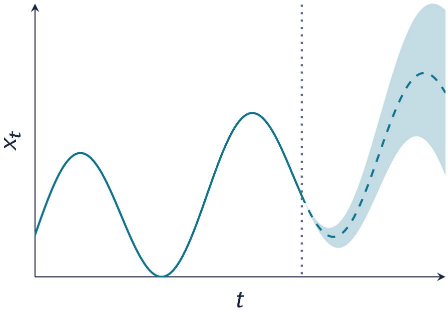
Given history $x_{1:T} \in \mathbb{R}^{T \times D}$ and covariates c, predict the next H steps.

- **Point forecast:** $\hat{x}_{T+1:T+H} = f(x_{1:T}, c)$
- **Probabilistic forecast:** model the full predictive distribution

$$p(x_{T+1:T+H} \mid x_{1:T}, c)$$

- Real data brings trend, multiple seasonalities, regime changes, missing values, and cold starts.

The whole course is a story about ever-richer models of this conditional distribution.



History, median forecast, and uncertainty fan

Baselines You Must Beat First

Statistical

Seasonal naive, ETS, ARIMA, Theta.

Fast, interpretable, shockingly hard to beat on single series.

Tree-based ML

Gradient boosting on lag and calendar features.

Won the M5 competition; still a production workhorse.

Early deep learning

DeepAR (RNN, probabilistic), N-BEATS / N-HiTS (MLP stacks).

Shine with many related series.

Lesson from the M-competitions: report a seasonal-naive and an ETS baseline next to every deep model. If you can't beat them, you don't have a model — you have overhead.

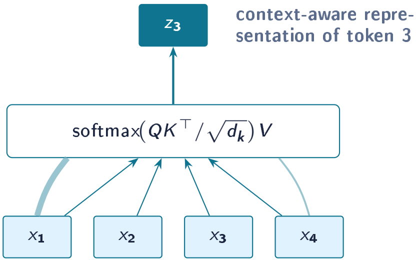
Transformer Basics I: Self-Attention

Each token emits a **query**, a **key**, and a **value**:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

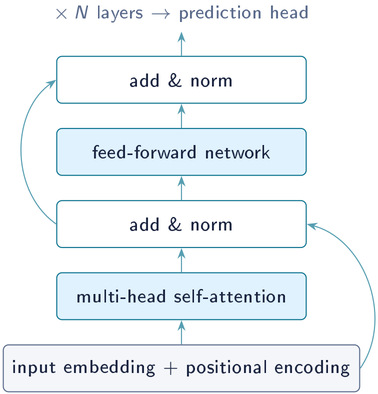
$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

- Every output is a *data-dependent weighted average* of all values.
- No recurrence: all pairs interact in one step — long-range dependencies become easy.
- Cost is $\mathcal{O}(L^2)$ in sequence length L — this will matter for long histories.



attention weights = learned relevance of each timestep

Transformer Basics II: The Full Block



- **Multi-head:** h attention maps in parallel subspaces; different heads can track different periodicities.
- **Positional encoding:** attention is permutation-invariant, so order must be injected — sinusoidal, learned, or rotary; for time series often enriched with calendar features (hour, weekday, holiday).
- **Residuals + normalization:** make deep stacks trainable.
- **Decoder-only vs encoder–decoder:** autoregressive generation vs direct multi-horizon output — both appear in forecasting models.

Open question for time series: what is a “token”? A single timestep? A patch? A variable? Each answer defines a model family.

Transformer Math I: One Head, Step by Step

Input sequence $X \in \mathbb{R}^{L\times d}$ (L tokens, model width d). Project into three roles:

$$Q = XW_Q, \quad K = XW_K \in \mathbb{R}^{L\times d_k}, \qquad V = XW_V \in \mathbb{R}^{L\times d_v}$$

1. Scores — compatibility of every query with every key:

$$S = \frac{QK^\top}{\sqrt{d_k}} \in \mathbb{R}^{L\times L}, \qquad S_{ij} = \frac{\mathrm{q}_i^\top \mathrm{k}_j}{\sqrt{d_k}}$$

2. Weights — row-wise softmax, so each row sums to 1:

$$A_{ij} = \frac{\exp(S_{ij})}{\sum_{j'=1}^{L} \exp(S_{ij'})}$$

3. Mix — each output is a convex combination of values:

$$\mathrm{z}_i = \sum_{j=1}^{L} A_{ij}\, \mathrm{v}_j, \qquad Z = AV$$

Why divide by $\sqrt{d_k}$?

If entries of q, k are unit-variance and independent, $\mathrm{q}^\top \mathrm{k}$ has variance d_k. Without scaling, large dot products push softmax into saturated regions where gradients vanish. Dividing by $\sqrt{d_k}$ keeps scores at unit scale.

Causal masking: for autoregressive forecasting, set $S_{ij} = -\infty$ for $j > i$ so a token attends only to its past.

Transformer Math II: Multi-Head and the Full Layer

Multi-head attention runs h independent heads in parallel and concatenates:

$$\mathrm{head}_m = \mathrm{Attn}(XW_Q^m, XW_K^m, XW_V^m), \qquad \mathrm{MHA}(X) = \left[\mathrm{head}_1 \,\|\, \cdots \,\|\, \mathrm{head}_h\right] W_O$$

Each head uses $d_k = d/h$, so total cost matches one full-width head while letting heads specialize (e.g. different seasonal lags).

One transformer layer (pre-norm form):

$$X' = X + \mathrm{MHA}\big(\mathrm{LN}(X)\big)$$
$$Y = X' + \mathrm{FFN}\big(\mathrm{LN}(X')\big)$$

with a position-wise feed-forward network

$$\mathrm{FFN}(\mathrm{u}) = W_2\,\sigma(W_1\mathrm{u} + \mathrm{b}_1) + \mathrm{b}_2$$

σ is GELU/ReLU; the hidden width is usually $4d$.

Sinusoidal positional encoding injects order:

$$\mathrm{PE}_{t,2i} = \sin\left(\frac{t}{10000^{2i/d}}\right), \;\; \mathrm{PE}_{t,2i+1} = \cos\left(\frac{t}{10000^{2i/d}}\right)$$

added to token embeddings so attention can read relative offsets.

Cost: scores S are $L \times L$, so compute and memory are $\mathcal{O}(L^2 d)$ — the bottleneck every long-horizon forecasting variant attacks.

Why Time Series Stress-Test the Transformer

- **Quadratic cost:** hourly data with a one-year context is $L \approx 8{,}760$ — L^2 attention gets expensive fast.
- **Weak tokens:** a single scalar timestep carries almost no semantics (unlike a word); attention over point-wise tokens is noisy.
- **Channel strategy:** mix variables in one token (channel mixing) or model each variate independently (channel independence)?
- **Distribution shift:** train/test statistics drift; normalization (e.g. RevIN) becomes a first-class design choice.

2019–2023 in one sentence: the field tried efficient attention first, got humbled by a linear model, then fixed the token definition.

From VAR to Attention

For anyone coming from classical multivariate time series, attention is a familiar idea in disguise.

Classical — VAR(p)

$$\mathrm{y}_t = \sum_{i=1}^{p} A_i\, \mathrm{y}_{t-i} + \varepsilon_t$$

- Coefficient matrices A_i are *fixed* — estimated once by OLS.
- Forecast is a *linear*, finite-lag combination of the past.

Modern — a transformer

$$\hat{\mathrm{y}}_t = \sum_{i=1}^{L} \underbrace{a_i(\mathrm{y}_{t-L:t-1})}_{\text{data-dependent}}\, v(\mathrm{y}_{t-i})$$

- Weights $a_i = \mathrm{softmax}(QK^\top/\sqrt{d_k})$ are *recomputed for every window*.
- Forecast is a *nonlinear* combination of learned value vectors.

One sentence: attention weights play the role of the VAR coefficients A_i — but they vary with the input instead of being fixed, and a causal mask makes attention strictly backward-looking like a VAR.

Reading the Model: IRF and Granger $\leftrightarrow$ Attention

Impulse response

Classical: $B_{ij}^{(h)}$ = effect on $y_{t+h}^{(i)}$ of a unit shock to $\varepsilon_t^{(j)}$, from the MA(∞) inversion.

Modern analog: attention / gradient-sensitivity maps show which past inputs drive a forecast — learned, nonlinear, horizon-dependent.

Granger causality

Classical: does x's past help predict y? F-test of $H_0 : b_1 = \cdots = b_p = 0$.

Modern analog: multivariate models (iTransformer) attend *across variables*; cross-variable weights measure predictive relevance.

Caveat worth teaching: attention is *associational*, not structural — it is not an orthogonalized shock and is not a hypothesis test. For causal effects use interventions or gradient sensitivities; for inference, keep the Granger F-test or add ablation importance.

The Efficiency Race (2019–2022)

Model	Key idea	Contribution
Informer AAAI 2021	ProbSparse attention: keep only the most informative queries	$\mathcal{O}(L \log L)$ attention; generative-style decoder for one-shot long-horizon output
Autoformer NeurIPS 2021	Series decomposition blocks + auto-correlation instead of dot-product	Builds trend/seasonal inductive bias directly into the architecture
FEDformer ICML 2022	Attention in the frequency domain (Fourier/wavelet bases)	Linear complexity; global spectral view of the series

All three beat vanilla transformers on the long-horizon benchmarks (ETT, Electricity, Traffic, Weather)...
which set the stage for an uncomfortable question.

2022: "Are Transformers Effective for Time Series Forecasting?"

DLinear (Zeng et al., AAAI 2023): decompose into trend + seasonal, fit *one linear layer* per component, map history directly to horizon.

- Outperformed Informer, Autoformer, and FEDformer on most long-horizon benchmarks.
- Diagnosis: point-wise tokens lose temporal semantics; iterated decoding accumulates error; benchmarks rewarded smooth extrapolation.
- Not the end of transformers — the end of *naive* transformers.

1 layer

linear model, no attention

$>$ 3 SOTA

transformers on 8/9 datasets

Takeaway for practitioners: complexity must earn its keep — always run the linear baseline.

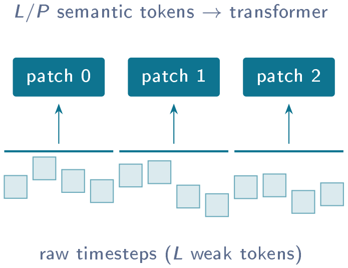
The Fix: Patches as Tokens

PatchTST (ICLR 2023)

- Split each series into patches of P steps; embed each patch as one token.
- Tokens become *meaningful local shapes*; context length shrinks by P, so attention cost drops by P^2.
- Channel independence: one shared backbone applied per variate.

iTransformer (ICLR 2024): invert the axes — treat each *variate* as a token and attend across variables.

Patching restored transformer SOTA — and became the input layer of nearly every foundation model in Module 3.



MODULE 3 1:50 – 2:45

Time Series Foundation Models

One pretrained model, any series · tokenizing time · the 2026 model zoo

The Zero-Shot Paradigm Shift

Before

One model per dataset (or per series).

Feature engineering, hyperparameter tuning, retraining pipelines.

Cold-start series are a hard problem.

After

One network pretrained on huge, diverse corpora (e.g. LOTSA: ∼27B observations) plus synthetic series.

Load it, feed history, get a probabilistic forecast — *zero-shot*.

Forecasting becomes model *selection*, not model *training*.

Direct analogy to LLMs — but time series add nonstationarity, mixed frequencies, irregular sampling, and very different value ranges across domains.

Tokenizing Time: Two Camps

Values as a vocabulary

Scale series, quantize values into discrete bins, train a language model on the tokens.

- Chronos-T5: forecasting literally *as* language modeling; probabilistic forecasts by sampling trajectories.
- Pro: reuses the whole LLM toolbox. Con: quantization error, long sequences.

Real-valued patch embeddings

PatchTST-style patches in, distribution parameters or quantiles out.

- TimesFM: decoder-only over patches; Moirai: any-variate attention, mixture-distribution head; Chronos-Bolt/Chronos-2: patches + direct quantile heads.
- Pro: compact, fast, native uncertainty. Now the dominant design.

Output heads to know: autoregressive sampling · direct quantiles · parametric mixtures · (new) generative flow/diffusion heads — a bridge to Module 4.

The Model Zoo (state of play, 2026)

Model	Org	Architecture	Size	Notable for
Chronos-2 (2025)	Amazon	encoder-only, patches	∼120M	Multivariate + covariates, in-context learning, strong zero-shot; successor to Chronos-T5/Bolt
TimesFM 2.5	Google	decoder-only, patches	∼200M	16k context, continuous quantile head, battle-tested in production
Moirai-2 (2026)	Salesforce	decoder-only	compact	Any-variate, messy real-world data focus; Moirai-MoE adds sparse experts
Toto	Datadog	decoder-only	∼150M	Observability/DevOps metrics specialization
TimeGPT	Nixtla	encoder–decoder	API	First commercial TSFM (2023); API-only
Lag-Llama	ServiceNow+	decoder-only	small	Lag features as tokens; open probabilistic baseline
Time-MoE / Sundial	THU et al.	MoE / flow head	up to 2.4B	Scaling studies; Sundial pairs a transformer with a generative flow-matching head

Open weights for most (Hugging Face); AutoGluon–TimeSeries and GluonTS wrap several behind one interface.

Do They Actually Work?

- On public leaderboards (GIFT-Eval, fev-bench), top TSFMs **zero-shot beat tuned statistical models and most task-specific deep models** on aggregate.
- Fine-tuning helps, but gains are often *modest and dataset-dependent* — try zero-shot first.
- Per-series inference is fast (hundreds of forecasts/sec on one GPU for small variants); small models run on CPU.

Caveats to teach your team: benchmark contamination is hard to rule out for pretrained models · covariate support is still uneven · irregular sampling and very long horizons remain weak spots · domain shift (e.g. finance) can erase the advantage — always backtest on *your* data.

Live Demo: Zero-Shot in a Few Lines

```
# pip install chronos-forecasting
import pandas as pd, torch
from chronos import BaseChronosPipeline

pipe = BaseChronosPipeline.from_pretrained(
    "amazon/chronos-2")

y = pd.read_csv("electricity.csv")["load"]
ctx = torch.tensor(y.values[-512:])

q, mean = pipe.predict_quantiles(
    context=ctx, prediction_length=48,
    quantile_levels=[0.1, 0.5, 0.9])
```

What to look at during the demo

- Forecast quality with *zero* training.
- Calibration of the 80% interval over a rolling backtest.
- Comparison against seasonal-naive and AutoETS on the same split.
- Runtime on CPU vs GPU.

We repeat this pattern in the Module 5 lab with your own data.

MODULE 4 2:55 – 3:55 (AFTER A 10-MINUTE BREAK)

Diffusion Models for Forecasting

Denoising math from scratch · TimeGrad · CSDI & SSSD · the generative frontier

Why Generative Models for Forecasting?

- A forecast is a **distribution over futures**, not a number. Decisions (inventory, trading, staffing) consume quantiles and scenarios.
- Quantile heads give marginal uncertainty per timestep, but **no coherent joint trajectories**: you cannot ask "what does a plausible bad week look like?"
- Generative models sample *whole future paths*: correlated across time and across variables — exactly what risk analysis, scenario planning, and downstream simulation need.
- Diffusion models are the current best-in-class generative family (images, audio, video) — Module 4 ports them to time series.

From Gaussian VARMA Densities to Sampled Scenarios

Classical — Gaussian VARMA

$$y_{t+h} \mid \mathcal{F}_t \sim \mathcal{N}(\hat{y}_{t+h}, \Sigma_h)$$

- Closed-form density, Σ_h built from the shock covariance W.
- But the shape is *fixed to be Normal* — no skew, no multi-modality.

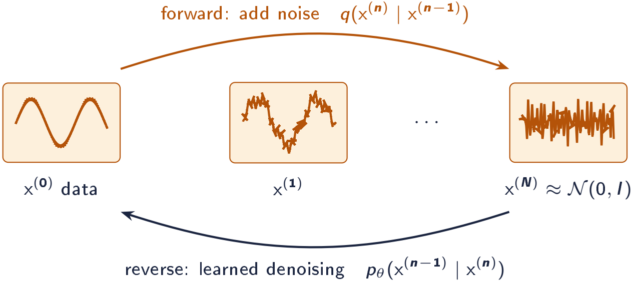
Modern — diffusion

$$y_{t+1:t+H} \sim p_\theta(\cdot \mid \mathcal{F}_t)$$

- *Learn to sample* whole future paths — no Gaussian assumption.
- Multi-modal, heavy-tailed, and *coherent* across time and variables.

Link: diffusion learns the joint predictive law that a Gaussian VARMA approximates with a single covariance Σ_h. **But** neither neural nor foundation models enforce $I(1)$ structure or a cointegrating equilibrium $\beta^\top y_t = 0$ — when a known long-run relation must hold, a classical **ECM** is still the right tool.

Diffusion Basics I: Destroy, Then Learn to Rebuild



- **Forward process (fixed):** $q(\mathrm{x}^{(n)} \mid \mathrm{x}^{(n-1)}) = \mathcal{N}\big(\sqrt{1-\beta_n}\,\mathrm{x}^{(n-1)}, \beta_n I\big)$ — gradually drown the signal in Gaussian noise.
- **Shortcut:** $\mathrm{x}^{(n)} = \sqrt{\bar{\alpha}_n}\,\mathrm{x}^{(0)} + \sqrt{1-\bar{\alpha}_n}\,\epsilon$, with $\bar{\alpha}_n = \prod_{i \le n}(1-\beta_i)$ — jump to any noise level in one step.
- **Generation:** start from pure noise, apply the learned reverse steps to synthesize new data.

Diffusion Basics II: Training and Sampling

Training reduces (after simplifying the ELBO) to noise prediction:

$$\mathcal{L}(\theta) \;=\; \mathbb{E}_{\mathrm{x}^{(0)},\, n,\, \epsilon\sim\mathcal{N}(0,I)} \left\| \epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_n}\,\mathrm{x}^{(0)} + \sqrt{1-\bar{\alpha}_n}\,\epsilon,\; n\right) \right\|^2$$

i.e. a denoising network learns "which noise was added?" at every corruption level. Equivalent view: it learns the *score* $\nabla_{\mathrm{x}} \log p(\mathrm{x})$ — the direction back toward the data manifold.

Sampling

- Iterate $\mathrm{x}^{(n)} \to \mathrm{x}^{(n-1)}$ using ϵ_θ; classic DDPM uses 50–1000 steps.
- Fast samplers (DDIM, DPM-Solver, distillation, flow matching) cut this to a handful of steps.

Conditioning — the key for forecasting

- Make the denoiser $\epsilon_\theta(\mathrm{x}^{(n)}, n, \mathrm{c})$ where c encodes the observed history (and covariates).
- Classifier-free guidance: train with and without c, interpolate at sampling time to sharpen conditional samples.

Forecasting as conditional generation: sample $\mathrm{x}_{T+1:T+H} \sim p_\theta(\cdot \mid \mathrm{x}_{1:T})$ many times $\Rightarrow$ an empirical predictive distribution of full trajectories.

Diffusion Math I: The Forward Process in Closed Form

A fixed Markov chain adds Gaussian noise over N steps with schedule $\beta_1, \ldots, \beta_N$:

$$q(\mathrm{x}^{(n)} \mid \mathrm{x}^{(n-1)}) = \mathcal{N}\big(\mathrm{x}^{(n)};\ \sqrt{1-\beta_n}\,\mathrm{x}^{(n-1)},\ \beta_n I\big)$$

Let $\alpha_n = 1 - \beta_n$ and $\bar{\alpha}_n = \prod_{i=1}^{n} \alpha_i$. Composing Gaussians gives a one-shot jump to any level:

$$\boxed{q(\mathrm{x}^{(n)} \mid \mathrm{x}^{(0)}) = \mathcal{N}\big(\sqrt{\bar{\alpha}_n}\,\mathrm{x}^{(0)},\ (1-\bar{\alpha}_n)I\big)}$$

$$\Rightarrow \quad \mathrm{x}^{(n)} = \sqrt{\bar{\alpha}_n}\,\mathrm{x}^{(0)} + \sqrt{1-\bar{\alpha}_n}\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

Why this matters

Training never simulates the whole chain: pick a random step n, draw one ϵ, and form $\mathrm{x}^{(n)}$ directly. As $n \to N$, $\bar{\alpha}_n \to 0$ and $\mathrm{x}^{(N)} \approx \mathcal{N}(0, I)$ — a tractable prior to start sampling from.

The true reverse step is also Gaussian when conditioned on $\mathrm{x}^{(0)}$: $q(\mathrm{x}^{(n-1)} \mid \mathrm{x}^{(n)}, \mathrm{x}^{(0)}) = \mathcal{N}(\tilde{\boldsymbol{\mu}}_n, \tilde{\beta}_n I)$ — the target the network learns to match.

Diffusion Math II: Reverse Process and Objective

The model learns the reverse chain with a denoising network ϵ_θ:

$$p_\theta(\mathrm{x}^{(n-1)} \mid \mathrm{x}^{(n)}) = \mathcal{N}\big(\mu_\theta(\mathrm{x}^{(n)}, n),\, \Sigma_n\big), \qquad \mu_\theta = \frac{1}{\sqrt{\alpha_n}}\Big(\mathrm{x}^{(n)} - \tfrac{\beta_n}{\sqrt{1-\bar{\alpha}_n}}\,\epsilon_\theta(\mathrm{x}^{(n)}, n)\Big)$$

Variational bound. Maximizing the data likelihood reduces to a sum of KL terms matching p_θ to the true Gaussian posteriors:

$$\mathcal{L}_{\mathrm{vlb}} = \mathbb{E}_q\Big[\textstyle\sum_n \mathrm{KL}\big(q(\mathrm{x}^{(n-1)} \,|\, \mathrm{x}^{(n)}, \mathrm{x}^{(0)}) \,\|\, p_\theta\big)\Big]$$

Simplified loss (DDPM). Reparameterizing collapses each KL to a plain regression on the injected noise:

$$\boxed{\mathcal{L} = \mathbb{E}_{\mathrm{x}^{(0)}, n, \epsilon}\Big\|\epsilon - \epsilon_\theta(\mathrm{x}^{(n)}, n)\Big\|^2}$$

Score-based view. Predicting noise is (up to scale) estimating the score:

$$\epsilon_\theta(\mathrm{x}^{(n)}, n) \approx -\sqrt{1-\bar{\alpha}_n}\,\nabla_{\mathrm{x}} \log q(\mathrm{x}^{(n)})$$

Sampling = following the score back to the data manifold, step by step.

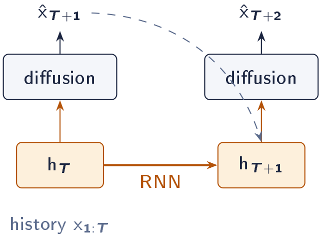
Sampling step: $\mathrm{x}^{(n-1)} = \mu_\theta(\mathrm{x}^{(n)}, n) + \sqrt{\Sigma_n}\,\mathrm{z}$, $\mathrm{z} \sim \mathcal{N}(0, I)$, from $n = N$ down to 1.

TimeGrad (2021): The First Diffusion Forecaster

- An RNN summarizes history into hidden state h_t; a **conditional diffusion head** models $p(x_{t+1} \mid h_t)$.
- Applied **autoregressively**: generate one multivariate step, feed it back, repeat over the horizon.
- Strong CRPS on high-dimensional multivariate benchmarks (e.g. thousands of correlated series).

Limitations

- Slow: a full reverse diffusion *per timestep*.
- Error accumulation over long horizons.



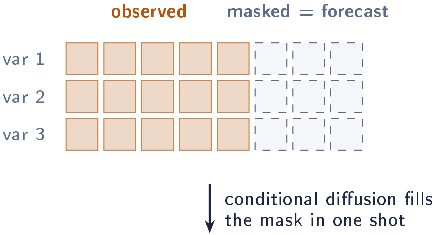
One reverse-diffusion sampling loop per forecast step, conditioned on the recurrent state.

CSDI & SSSD: Forecasting as Imputation

CSDI (NeurIPS 2021)

- Treat the future as **missing values**: mask it, condition the diffusion on the observed part, generate the whole masked block *non-autoregressively*.
- Two-dimensional attention over time and features; one model serves imputation *and* forecasting.

SSSD (2022): same recipe, but transformer layers replaced by **S4 state-space layers** — cheaper on long sequences.



No step-by-step decoding ⇒ no error accumulation; handles arbitrary missingness patterns for free.

Sharper Conditioning: TimeDiff and TSDiff

TimeDiff (ICML 2023)

Non-autoregressive, with *time-series-aware* conditioning:

Future mixup — blend ground-truth future into the condition during training.

AR initialization — start the reverse process from a cheap linear forecast instead of pure noise.

Faster and more accurate than masking-style designs on long horizons.

TSDiff (NeurIPS 2023)

One *unconditional* diffusion model per domain.

Self-guidance at inference conditions it on any observation pattern — forecasting, imputation, refinement — without retraining.

Doubles as a generator of **synthetic training data** for downstream models.

Design axes to remember: autoregressive vs whole-segment · where conditioning enters · data space vs latent space · how many sampling steps you can afford.

The Generative Frontier (2024–2026)

- **Multi-scale & decomposition:** MG-TSD, mr-Diff — denoise coarse-to-fine.
- **Non-stationary diffusion:** NsDiff (ICML 2025) adapts the noise process to drifting variance.
- **Latent & multimodal:** diffuse in a learned latent space; condition on text or vision (LDM4TS, MCD-TSF).
- **Few-step sampling:** DDIM/DPM-Solver, distillation, and **flow matching** make generation cheap enough for production.
- **Convergence with Module 3:** generative heads inside foundation models (e.g. Sundial's flow-matching head) — pretrained backbone, sampled trajectories.

Synthesis: transformers solved *representation*; foundation models solved *generalization*; diffusion solves *uncertainty*. The frontier is all three in one model.

MODULE 5 3:55 – 4:45

Hands-On & Practice

Lab · honest evaluation · choosing a model · open problems

Hands-On Lab — Six Exercises (25 min)

1. **Build the data** — synthetic hourly load, or load your own CSV; train/test split.
2. **Classical VAR** — simulate a VAR(1), recover A by OLS, run a Granger F-test, plot the IRF.
3. **Baselines & MASE** — seasonal-naive with a backtest-based 80% interval; add AutoETS.
4. **Self-attention from scratch** — 12 lines of numpy; add a causal mask and multiple heads.
5. **Diffusion forward process** — noise a window with the closed-form q; stub the reverse step.
6. **Probabilistic scoring** — sample paths, CRPS, coverage; then beat it with Chronos-2 zero-shot.

Everything runs on **CPU** with `numpy` + `matplotlib` (`exercises.py`). The zero-shot stretch goal adds `pip install chronos-forecasting gluonts statsforecast`.

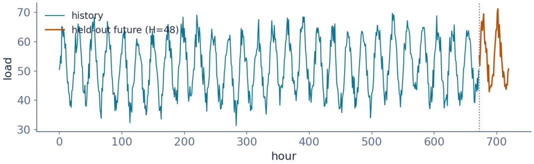
Exercise 1 — Build the Dataset

```python
import numpy as np

def make_series(n=24*30, seed=2):
    rng = np.random.default_rng(seed)
    t = np.arange(n)
    trend  = 50 + 0.005*t
    daily  = 12*np.sin(2*np.pi*t/24)
    weekly =  2*np.sin(2*np.pi*t/(24*7))
    noise  = rng.normal(0, 2.5, n)
    return t, trend+daily+weekly+noise

t, y = make_series()
H = 48                          # horizon
y_train, y_test = y[:-H], y[-H:]
```

Your turn: swap in a real CSV (`pd.read_csv`); identify the dominant seasonal period with an autocorrelation plot.



Hourly synthetic load: trend + daily + weekly + noise. The last 48 hours (amber) are held out for every model in this lab.

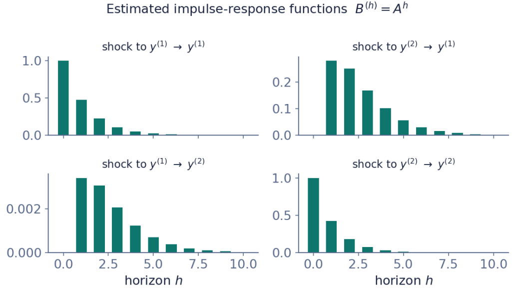
Exercise 2 — Classical VAR(1)

```
A_true = np.array([[0.5, 0.3],
                   [0.0, 0.4]])    # y2 not driven
                                     by y1

def simulate_var1(A, T=400, seed=3):
    rng = np.random.default_rng(seed)
    Y = np.zeros((T, A.shape[0]))
    for s in range(1, T):
        Y[s] = A@Y[s-1] + rng.normal(0,1,A.shape
            [0])
    return Y

def fit_var1(Y):                    # OLS: Y_t = A Y_
     {t-1}
    X, Z = Y[1:], Y[:-1]
    return np.linalg.lstsq(Z, X, rcond=None)[0].T

Y = simulate_var1(A_true)
A_hat = fit_var1(Y)                 # ~ A_true
B = [np.linalg.matrix_power(A_hat, h)
     for h in range(11)]            # IRF: B_h = A^h
```



Your turn: implement the Granger F-test of H_0 : lags of y_1 don't enter the y_2 equation. You should find $F_{y_1 \to y_2} \approx 0$ but $F_{y_2 \to y_1} \gg 0$.

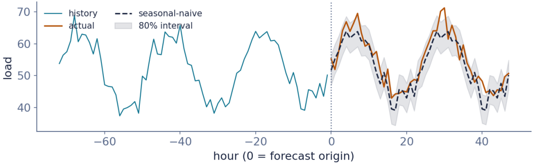
Exercise 3 — Baselines and MASE

```
m = 24                          # daily period
sn = y_train[-m:][np.arange(H) % m]

# rolling-origin backtest -> honest spread
def bt_resid(y, m, H, n=120):
    R = []
    for o in range(len(y)-H-n, len(y)-H):
        f = y[o-m:o][np.arange(H) % m]
        R.append(y[o:o+H] - f)
    return np.array(R)

sig = bt_resid(y_train, m, H).std(0)
lo, hi = sn - 1.28*sig, sn + 1.28*sig    # 80%

mase = (np.abs(y_test - sn).mean() /
        np.abs(y_train[m:]-y_train[:-m]).mean())
print(mase)    # ~1.06
```



Your turn: add an AutoETS forecast (`statsforecast`); report MASE for both. Why build the interval from *backtest* residuals, not one-step errors?

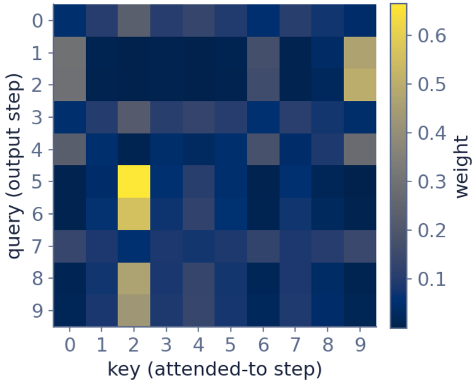
Exercise 4 — Self-Attention from Scratch

```
def softmax(x, axis=-1):
    x = x - x.max(axis, keepdims=True)
    e = np.exp(x)
    return e / e.sum(axis, keepdims=True)

def attention(X, Wq, Wk, Wv):
    Q, K, V = X@Wq, X@Wk, X@Wv
    dk = Q.shape[-1]
    S = Q @ K.T / np.sqrt(dk)       # scores
    A = softmax(S)                  # weights
    return A @ V, A                 # output, map

Z, A = attention(X, Wq, Wk, Wv)
```

Your turn: (1) add a causal mask ($S_{ij} = -\infty$ for $j > i$); (2) split into h heads and concatenate. Confirm each row of `A` sums to 1.



Attention matrix A over 10 daily tokens: bright cells show which past days each output step leans on.

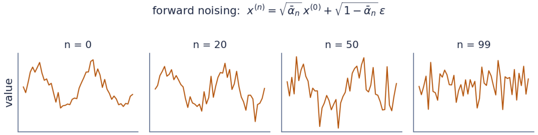
Exercise 5 — Diffusion Forward Process

```
N = 100
betas  = np.linspace(1e-4, 0.08, N)
abar   = np.cumprod(1 - betas)    # alpha-bar

def q_sample(x0, n, rng):
    eps = rng.standard_normal(len(x0))
    return (np.sqrt(abar[n])*x0 +
            np.sqrt(1-abar[n])*eps)

x0 = (y_test - y_test.mean())/y_test.std()
rng = np.random.default_rng(0)
xn = {n: q_sample(x0, n, rng)
      for n in (0, 20, 50, 99)}
```

Your turn: given a trained `eps_theta(x,n)`, implement one reverse step $x^{(n-1)} = \frac{1}{\sqrt{\alpha_n}}(x^{(n)} - \frac{\beta_n}{\sqrt{1-\bar{\alpha}_n}}\,\epsilon_\theta) + \sqrt{\beta_n}z$ and sample from noise.



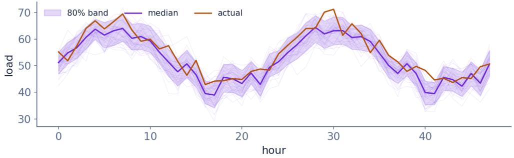
A clean window dissolves into Gaussian noise as $n \to N$. Training asks a network to invert one step of this at a random n.

Exercise 6 — Probabilistic Forecast & Scoring

```python
# 200 correlated sample paths (AR(1) noise)
S, phi = 200, 0.6
ar = np.zeros((S, H)); g = np.random.default_rng
    (7)
ar[:,0] = g.normal(0, sig[0], S)
for h in range(1, H):
    e = g.normal(0, sig[h]*np.sqrt(1-phi**2), S)
    ar[:,h] = phi*ar[:,h-1] + e
samples = sn[None,:] + ar
q10,q50,q90 = np.percentile(samples,[10,50,90],0)

def crps(samps, obs):            # sample CRPS
    a = np.abs(samps-obs).mean(0)
    b = np.abs(samps[:,None]-samps[None]).mean
        ((0,1))
    return (a - 0.5*b).mean()

cov = np.mean((y_test>=q10)&(y_test<=q90))
print(crps(samples, y_test), cov)  # ~2.3, ~0.7
```



Your turn: an 80% band covering ∼70% is mildly *under-calibrated* — diagnose and fix. Then beat this CRPS with Chronos-2 zero-shot.

Evaluating Forecasts Without Fooling Yourself

Metrics

- **Point:** MASE (scale-free, robust), sMAPE; never plain MAPE near zeros.
- **Probabilistic:** CRPS, weighted quantile loss, interval coverage — a sharp but miscalibrated model is a liability.

$$\mathrm{CRPS}(F, y) = \int \left(F(z) - \mathbb{1}\{y \leq z\}\right)^2 dz$$

Backtesting traps

- Use **rolling-origin** evaluation, not one fixed split.
- Fit scalers/normalizers on the training window only.
- Beware overlapping windows inflating sample counts.
- For TSFMs: your "test set" may resemble their pretraining data — validate on genuinely private series.

Choosing a Model in Practice

Your situation	Reach for
A few series, interpretability matters	ETS / ARIMA / Theta
Many related series, rich covariates, tabular culture	Gradient boosting or PatchTST-class model
Cold start, no ML team, fast time-to-value	Foundation model, zero-shot (Chronos-2, TimesFM)
Risk decisions need coherent scenario paths	Diffusion forecaster, or a TSFM with a generative head
Tight latency / edge deployment	Linear (DLinear) or distilled small model

Default workflow: seasonal-naive $\rightarrow$ statistical baseline $\rightarrow$ zero-shot TSFM $\rightarrow$ fine-tune or go generative only if the backtest says the gap is worth it.

Open Problems & Where the Field Is Going

- **Irregular & multimodal data:** events, text, and exogenous signals fused with numeric series.
- **Very long horizons** and hierarchical/grouped coherence constraints.
- **Calibration under drift:** guarantees when the world changes mid-deployment.
- **Benchmark hygiene:** contamination-free evaluation for pretrained models.
- **Agentic pipelines:** LLM agents that select, run, and monitor forecasting models end-to-end.

References I — Classical, Transformers, Foundation Models

Classical multivariate time series

- Sims, *Macroeconomics and Reality*, Econometrica 1980.
- Granger, *Investigating Causal Relations*, Econometrica 1969.
- Engle & Granger, *Co-integration and Error Correction*, Econometrica 1987.
- Johansen, *Estimation and Testing of Cointegration Vectors*, Econometrica 1991.
- Lütkepohl, *New Introduction to Multiple Time Series*, 2005; Hamilton, *Time Series Analysis*, 1994.

Transformers for forecasting

- Vaswani et al., *Attention Is All You Need*, NeurIPS 2017.
- Zhou et al., *Informer*, AAAI 2021; Wu et al., *Autoformer*, NeurIPS 2021; Zhou et al., *FEDformer*, ICML 2022.

Transformers (cont.)

- Zeng et al., *Are Transformers Effective?* (DLinear), AAAI 2023.
- Nie et al., *PatchTST*, ICLR 2023; Liu et al., *iTransformer*, ICLR 2024.
- Salinas et al., *DeepAR*, IJF 2020; Oreshkin et al., *N-BEATS*, ICLR 2020; Kim et al., *RevIN*, ICLR 2022.

Foundation models

- Ansari et al., *Chronos*, TMLR 2024; Das et al., *TimesFM*, ICML 2024; Woo et al., *Moirai* & LOTSA, ICML 2024.
- Garza & Mergenthaler-Canseco, *TimeGPT-1*, 2023; Rasul et al., *Lag-Llama*, 2023; Shi et al., *Time-MoE*, ICLR 2025.
- Aksu et al., *GIFT-Eval*, 2024.

References II — Diffusion, Surveys, Evaluation

Diffusion & score-based models

- Ho et al., *DDPM*, NeurIPS 2020; Song et al., *Score-based SDEs*, ICLR 2021.
- Song et al., *DDIM*, ICLR 2021; Ho & Salimans, *Classifier-Free Guidance*, 2022; Lipman et al., *Flow Matching*, ICLR 2023.
- Gu et al., *S4 (structured state spaces)*, ICLR 2022.

Diffusion for forecasting

- Rasul et al., *TimeGrad*, ICML 2021; Tashiro et al., *CSDI*, NeurIPS 2021.
- Alcaraz & Strodthoff, *SSSD*, TMLR 2023; Shen & Kwok, *TimeDiff*, ICML 2023.

Diffusion for forecasting (cont.)

- Kollovieh et al., *TSDiff*, NeurIPS 2023.
- Fan et al., *MG-TSD*, ICLR 2024; Shen et al., *mr-Diff*, ICLR 2024.

Surveys & evaluation

- Yang et al., *Survey on Diffusion Models for Time Series*, 2024.
- Meijer & Chen, *The Rise of Diffusion Models in TS Forecasting*, 2024.
- Makridakis et al., *M4*, IJF 2020; *M5*, IJF 2022.
- Gneiting & Raftery, *Strictly Proper Scoring Rules*, JASA 2007.

Full BibTeX in `references.bib`; typeset list in `references.pdf`.

Thank You

Core reading

Vaswani et al. 2017 · Zhou et al. (Informer) 2021 · Zeng et al. (DLinear) 2023 · Nie et al. (PatchTST) 2023

Ansari et al. (Chronos) 2024 · Das et al. (TimesFM) 2024 · Woo et al. (Moirai) 2024

Rasul et al. (TimeGrad) 2021 · Tashiro et al. (CSDI) 2021 · Shen & Kwok (TimeDiff) 2023 · Kollovieh et al. (TSDiff) 2023

Su et al. (Diffusion-for-TSF survey) 2025 · Meijer & Chen (survey) 2024

Slides, outline, and lab notebook accompany this deck.